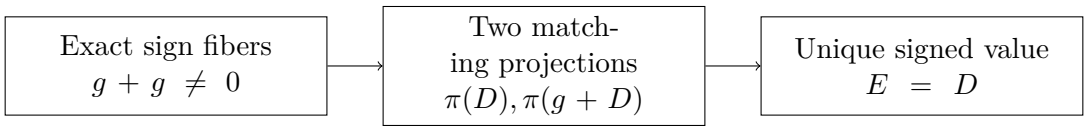
Independent conditional signed-pair review

1 November 2026

Three producer declarations, three producer controls and eight reviewer controls passed fresh isolated Lean replay. A separate axiom readback covered all 14 declarations and listed only propext, Classical.choice and Quot.sound. No discrepancy was found in the abstract conditional uniqueness statement.



Independent derivation. Equal first projections imply $E = D$ or $E = -D$. In the opposite-sign case the neighbor equality either implies equality directly or forces $g = -g$, contradicting $g + g \neq 0$. This establishes uniqueness, not existence. The prospective plan preceded proof-body inspection.

Reviewer controls. Actual sign-quotient functions on ZMod4 and ZMod5 have kernel-checked fibers. Eight controls cover their fiber identities, a nonzero order-two anchor ambiguity, insufficiency of one projection, neighbor separation, failure of a constant projection's fiber property, application of the main theorem, and an empty candidate set for an impossible requested coordinate. No native_decide was used.

Source correspondence. The saved delta_lift and degree-one boundary_lift filter candidates using validity and both projection matches, returning a value only when exactly one remains. This matches the theorem's conditional uniqueness interface. The identity shortcut instead recognizes a chart coordinate pattern and checks only the anchor-neighbor projection; that coordinate recognition needs its own correctness statement.

Projective boundary. Raw same_k checks cross-product equality, which accepts the zero vector against every vector. The reviewed filters reject zero supplied vectors and project_any rejects zero output. Connecting these checks to the abstract quotient function still requires the nonzero-representative invariant, projective quotient construction and exact source fiber theorem. The anchor premise also remains a caller obligation.

Trust and scope. Fresh private target modules used Lean4.34.1 and retained dependency caches. The adjacent independent-product source/object snapshot was hash-bound around replay; broader Mathlib caches were reused. Inputs are private exact snapshots, not published commits. Full formal-review workflow completion, compiled Rust equivalence, geometric fiber correctness and xadd closure are not claimed.

Source anchors: source/pruned.rs lines150–181; source/walk_impl_uncached.rs lines51–53. Exact hashes, replay receipts, full controls and source findings accompany REVIEW.md and status.json. Full cohort scope remains 1,260; no scientific execution or measurements occurred.